

VIOOLSDRIF, South Africa

WMO: 684110

Lat: 28.769S

Lon: 17.623E

Elev: 174

StdP: 99.25

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	10.2	-11.5	1.4	24.2	-9.6	1.7	25.4	7.2	20.1	6.3	19.4	1.1	320	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.7	41.5	19.4	40.0	19.1	38.6	18.8	23.8	31.9	22.5	32.6	21.4	33.1	2.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	16.1	30.6	18.9	14.0	28.9	17.2	12.6	27.8	72.3	31.9	66.8	32.9	62.9	33.0	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.7	6.0	DB	6.1	45.2	2.1	1.5	4.6	46.3	3.4	47.2	2.3	48.1	0.8	49.2	(4)
(5)			WB	2.7	23.3	1.3	0.7	1.8	23.8	1.0	24.2	0.2	24.6	-0.7	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.6	30.3	30.6	29.5	26.2	21.6	17.6	17.6	18.8	21.7	25.2	27.1	28.9	(6)	
	DBStd	5.94	3.44	3.27	3.65	4.01	3.61	2.60	3.08	3.31	3.79	4.28	4.12	3.49	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	149	0	0	0	1	11	44	52	33	8	1	0	0	(9)	
	CDD10.0	5312	629	578	604	487	360	227	234	274	351	471	512	586	(10)	
	CDD18.3	2420	371	344	346	237	112	21	28	48	109	213	262	328	(11)	
	CDH23.3	28437	4761	4484	4340	2672	1008	203	284	508	1056	2252	2943	3926	(12)	
	CDH26.7	15822	2901	2746	2555	1377	392	44	64	169	471	1185	1629	2289	(13)	
(14) Wind	WSAvg	2.4	2.8	2.6	2.3	2.3	1.9	1.9	1.9	2.2	2.5	2.8	2.9	2.9	(14)	
(15) Precipitation	PrecAvg	62	6	9	9	9	5	5	4	5	1	3	2	2	(15)	
	PrecMax	108	21	42	38	48	24	13	18	22	9	19	9	9	(16)	
	PrecMin	26	1	2	1	0	0	0	0	0	0	0	0	1	(17)	
	PrecStd	23	5	10	8	13	6	4	4	6	2	5	2	2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.6	42.7	41.9	39.3	35.6	30.3	31.0	35.1	36.9	40.7	41.3	42.6	(19)	
		MCWB	19.6	20.0	19.9	19.0	16.7	14.5	14.6	15.0	16.3	18.4	18.3	19.5	(20)	
	2%	DB	41.0	40.6	39.9	37.0	32.7	27.8	28.3	30.6	34.2	37.7	38.8	40.1	(21)	
		MCWB	19.3	19.7	19.7	18.1	15.5	13.4	13.3	13.5	15.0	17.4	17.5	19.0	(22)	
	5%	DB	39.1	39.0	38.0	35.1	30.6	25.6	26.3	28.2	31.9	35.5	36.8	38.0	(23)	
		MCWB	19.3	19.3	19.1	17.5	15.2	12.8	12.6	12.9	14.3	16.6	17.0	18.5	(24)	
	10%	DB	37.1	37.2	36.2	33.2	28.7	23.6	24.5	26.1	29.4	33.1	34.7	36.0	(25)	
		MCWB	18.9	19.2	18.6	17.0	14.7	12.3	12.0	12.2	13.6	15.9	16.4	17.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	24.7	25.5	21.0	20.7	15.7	15.1	15.7	17.3	22.1	21.1	24.8	(27)	
		MCDB	33.2	31.7	35.7	30.3	27.2	26.0	28.7	30.4	33.8	34.0	33.3	34.8	(28)	
	2%	WB	23.1	23.0	23.2	20.1	18.5	14.6	14.0	14.7	16.0	19.9	19.1	22.2	(29)	
		MCDB	32.6	31.6	31.1	31.9	25.7	23.6	26.0	27.6	32.2	32.4	32.3	32.6	(30)	
	5%	WB	21.4	22.1	21.8	19.3	17.0	13.8	13.2	13.7	15.1	18.2	18.0	20.5	(31)	
		MCDB	34.2	32.9	32.9	31.0	26.6	22.5	24.0	25.3	29.6	31.7	32.7	33.1	(32)	
	10%	WB	20.2	21.2	20.4	18.4	15.7	13.2	12.7	12.9	14.2	17.0	17.2	19.0	(33)	
		MCDB	34.2	32.9	33.4	30.5	25.8	21.5	22.7	24.0	27.6	30.5	31.5	33.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	12.7	12.4	11.4	11.6	11.5	12.2	12.2	12.0	12.7	13.0	13.6	(35)	
		MCDBR	14.9	13.9	13.7	13.3	14.2	14.6	15.0	15.5	15.6	15.5	15.3	15.5	(36)	
	5% WB	MCWBR	3.9	4.1	4.1	4.4	4.9	5.6	6.0	6.1	5.7	5.2	4.5	4.1	(37)	
		MCWBR	11.8	11.0	11.6	11.1	11.9	11.7	13.2	13.9	14.1	14.3	14.7	14.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.350	0.364	0.349	0.324	0.300	0.289	0.295	0.322	0.347	0.356	0.330	0.334	(40)	
		taud	2.459	2.444	2.470	2.528	2.559	2.550	2.520	2.422	2.329	2.335	2.462	2.474	(41)	
	Ebn at Noon		991	959	942	919	898	887	894	906	928	956	1004	1010	(42)	
		Edh at Noon	120	118	109	93	81	77	83	101	123	130	119	119	(43)	
(44) All-Sky Solar Radiation	RadAvg		8.57	7.79	6.70	5.28	4.16	3.61	3.88	4.78	6.06	7.39	8.37	8.88	(44)	
	RadStd		0.29	0.26	0.19	0.17	0.14	0.11	0.13	0.17	0.21	0.19	0.24	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page